MobaXtrem

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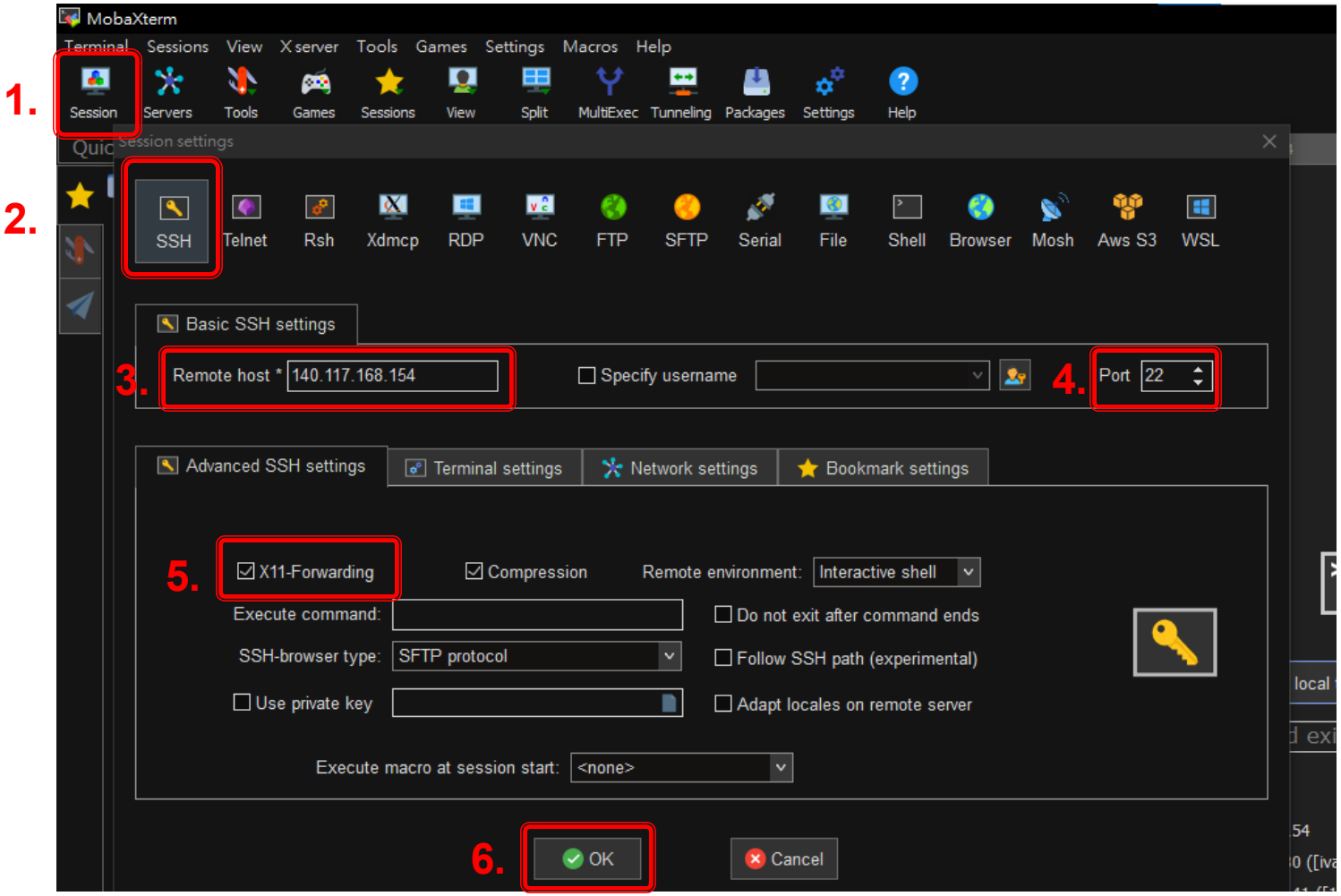
2025 ALU / HDL

X Tools

- 請使用遠端程式 MobaXterm 連入伺服器(以方便使用圖形化介面)

- 下載網址：https://mobaxterm.mobatek.net/download-home-edition.html

SSH

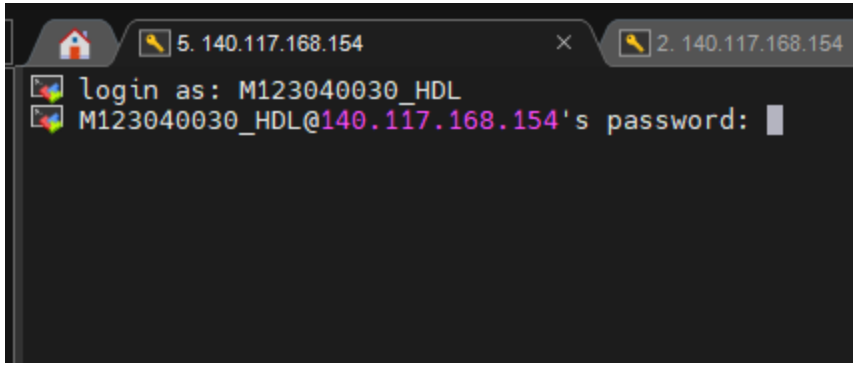

1.
2.
3.
4.
5.
6.
MobaXterm
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help
Session settings
SSH Telnet Rsh Xdmcp RDP VNC FTP SFTP Serial File Shell Browser Mosh Aws S3 WSL
Basic SSH settings
Remote host * 140.117.168.154
Specify username
Port 22
Advanced SSH settings
Terminal settings
Network settings
Bookmark settings
X11-Forwarding
Compression
Remote environment: Interactive shell
Execute command:
Do not exit after command ends
SSH-browser type: SFTP protocol
Follow SSH path (experimental)
Use private key
Adapt locales on remote server
Execute macro at session start: <none>
OK
Cancel

Login

- Username：
  - <Student ID>_HDL
  - <Student ID>_ALU

  Ex: M123040033_ALU
- Password：
  - 2025<Student ID>

  Ex: 2025M123040033



*輸入密碼時畫面不會顯示任何字元
看起來像沒反應是正常現象
可以連續按back以重新輸入

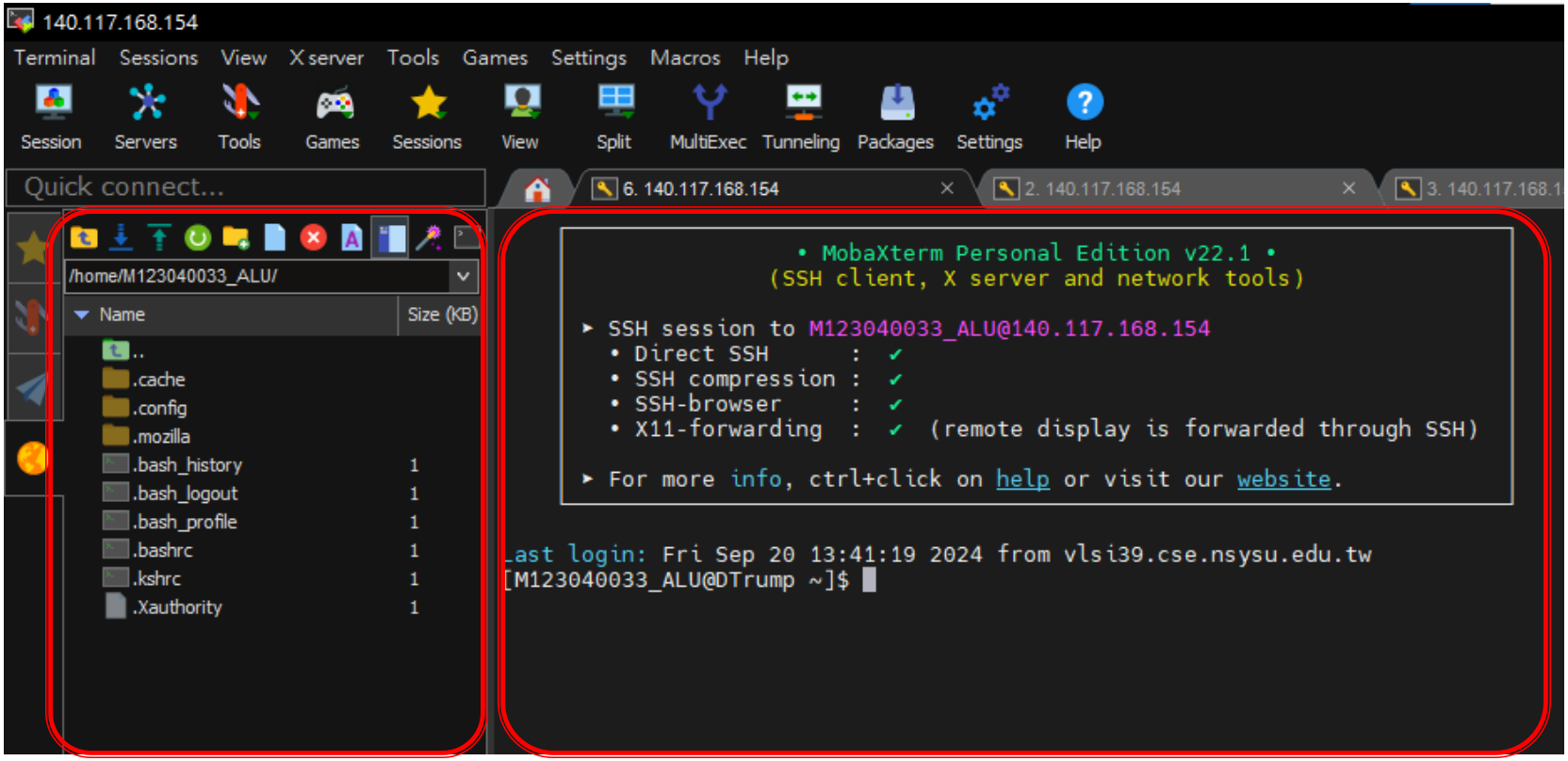
Security

為避免外部惡意攻擊，作業用伺服器有兩層防護：

1. 僅允許校內網域聯入。
   在校外請使用本校 VPN；
   外校生請額外告知。

2. 3分鐘內連線失敗(輸入錯誤)五次，
   將會禁止連線10分鐘。

登入成功畫面



檔案

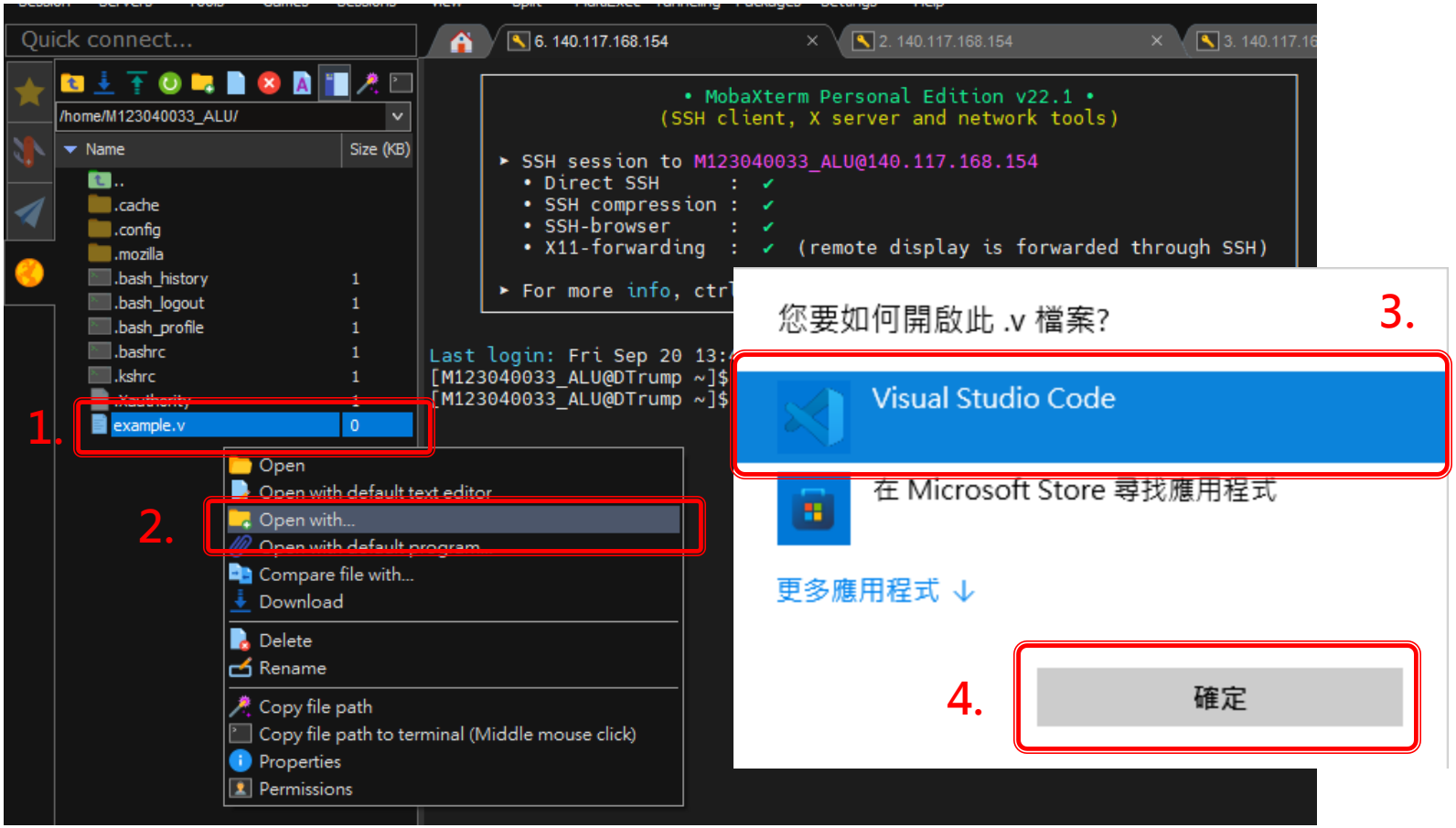
終端機

常用指令

- touch example.v //新增檔案
- mv example.v Homework1/. //移動/修改檔名
- cp example.v example2.v //複製檔案
- passwd //更改密碼
- 請自行更改密碼，防止他人登入抄襲作業

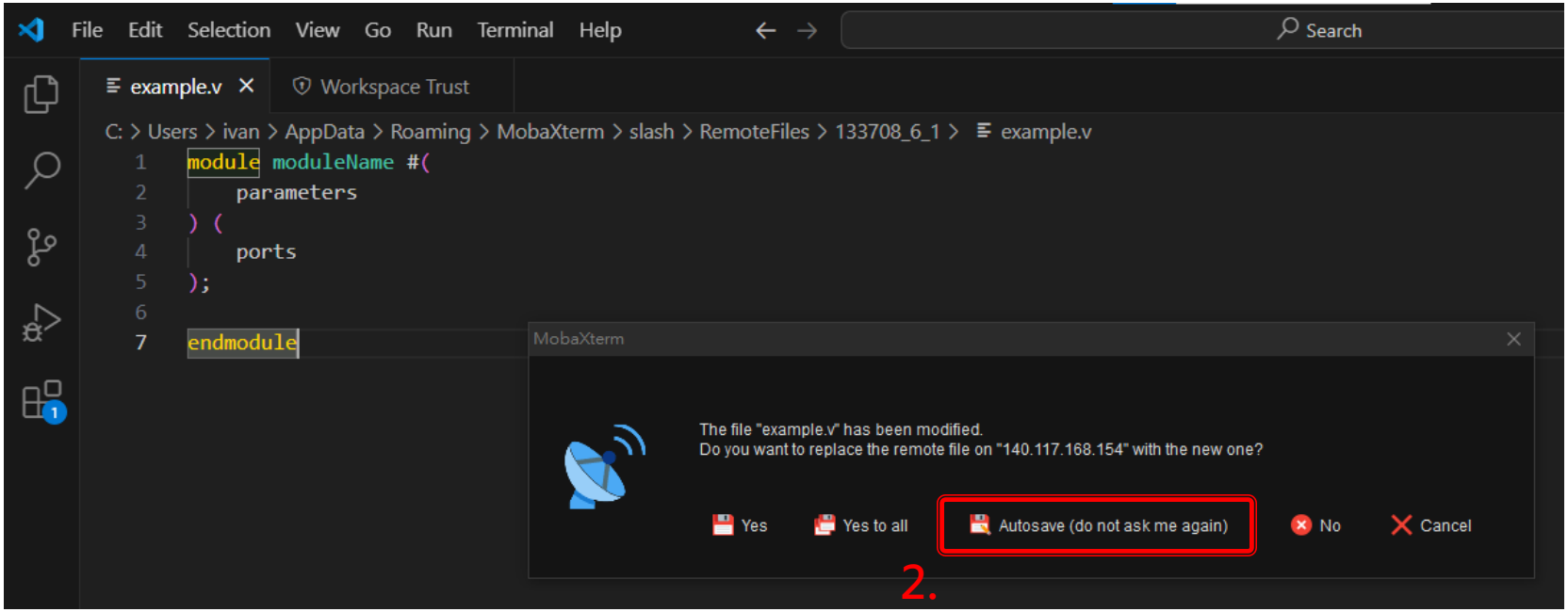
編輯檔案

- 可以使用自己電腦上的文字編輯器編寫程式碼



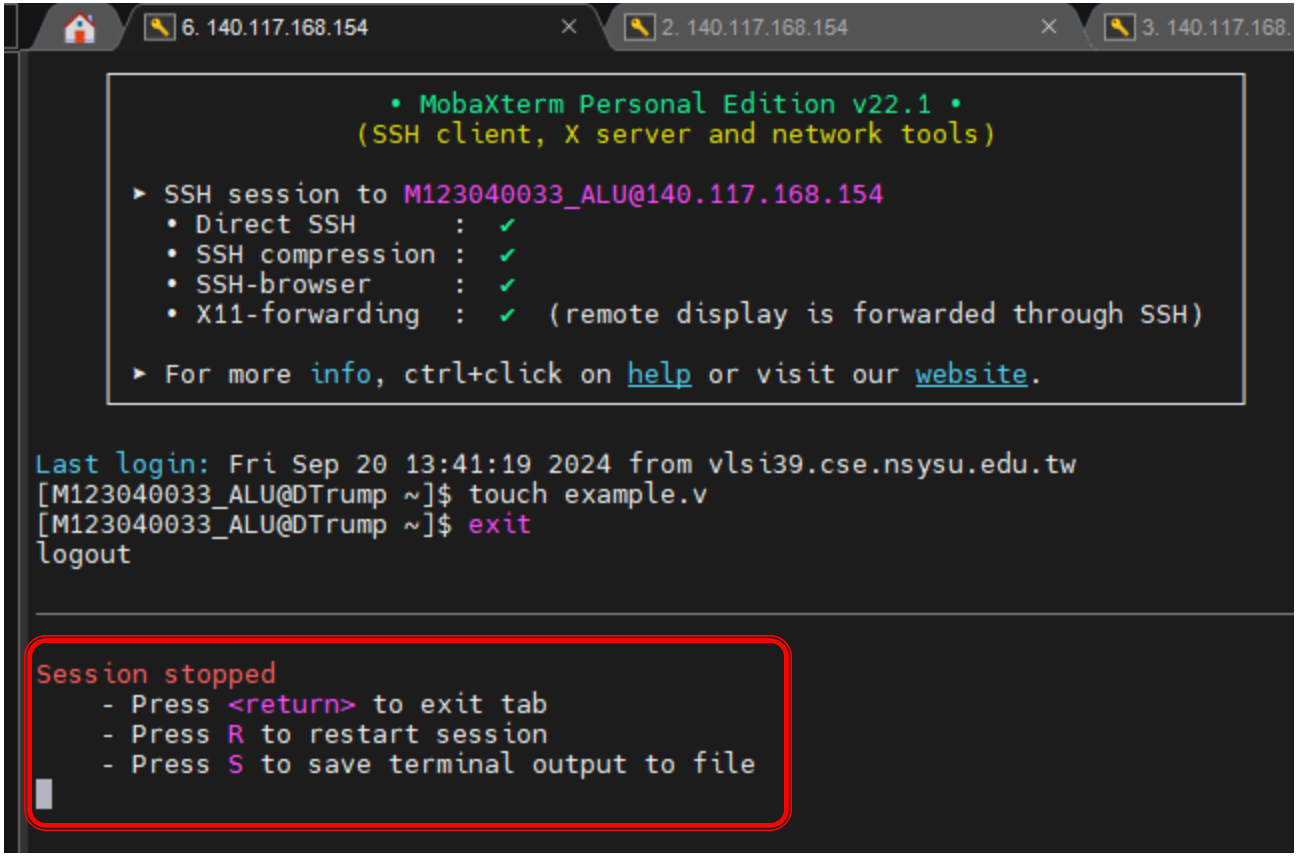
儲存檔案

1. Ctrl+S


File Edit Selection View Go Run Terminal Help
Search
example.v
Workspace Trust
C: > Users > ivan > AppData > Roaming > MobaXterm > slash > RemoteFiles > 133708_6_1 > example.v
module moduleName #(
parameters
) (
ports
);
endmodule
MobaXterm
The file "example.v" has been modified.
Do you want to replace the remote file on "140.117.168.154" with the new one?
Yes
Yes to all
Autosave (do not ask me again)
No
Cancel
2.

關閉連線

- 使用完畢後請務必輸入 `exit` / `(Ctrl + D)` 關閉連線以避免佔用伺服器資源



Homework1說明

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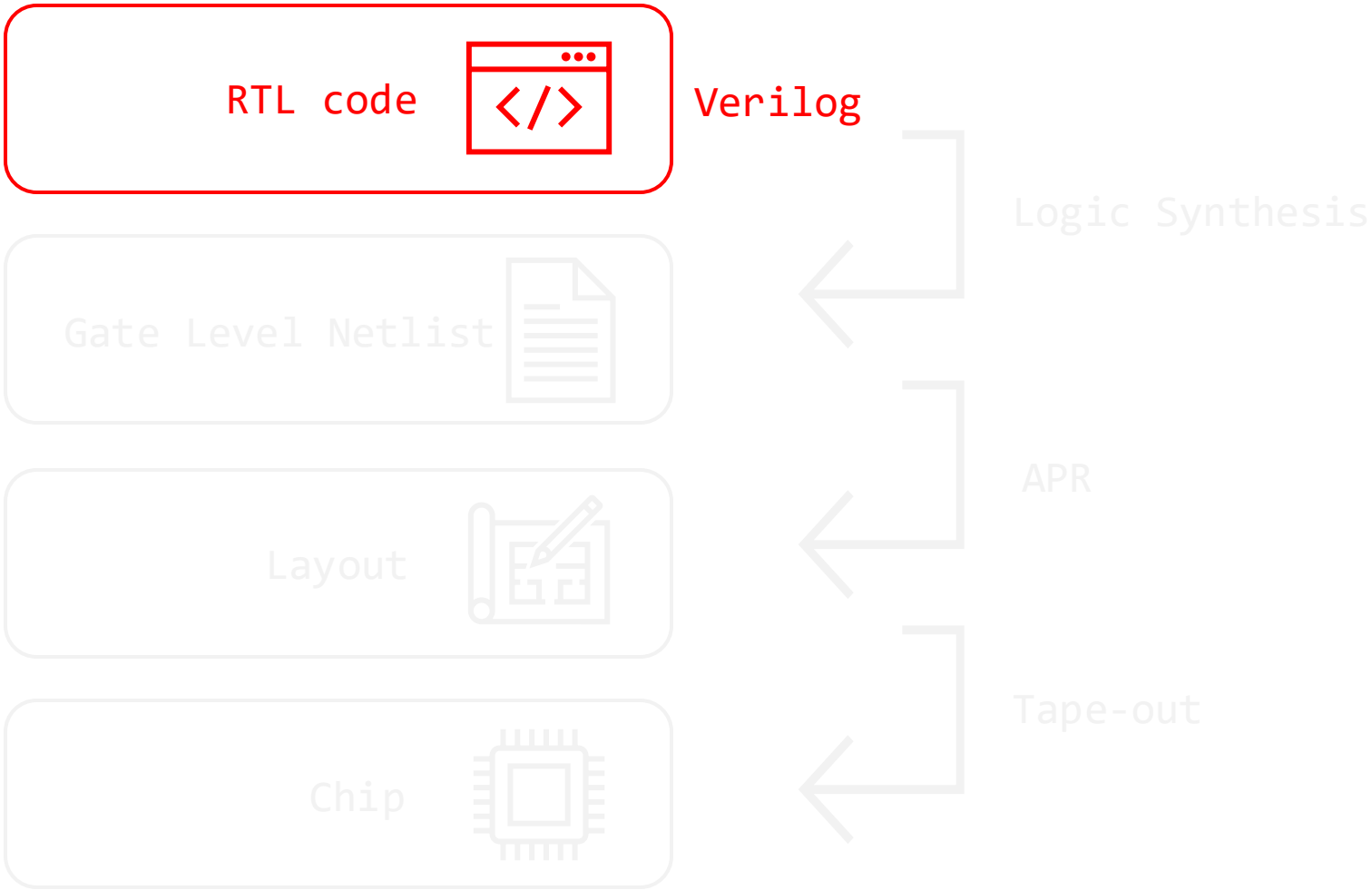
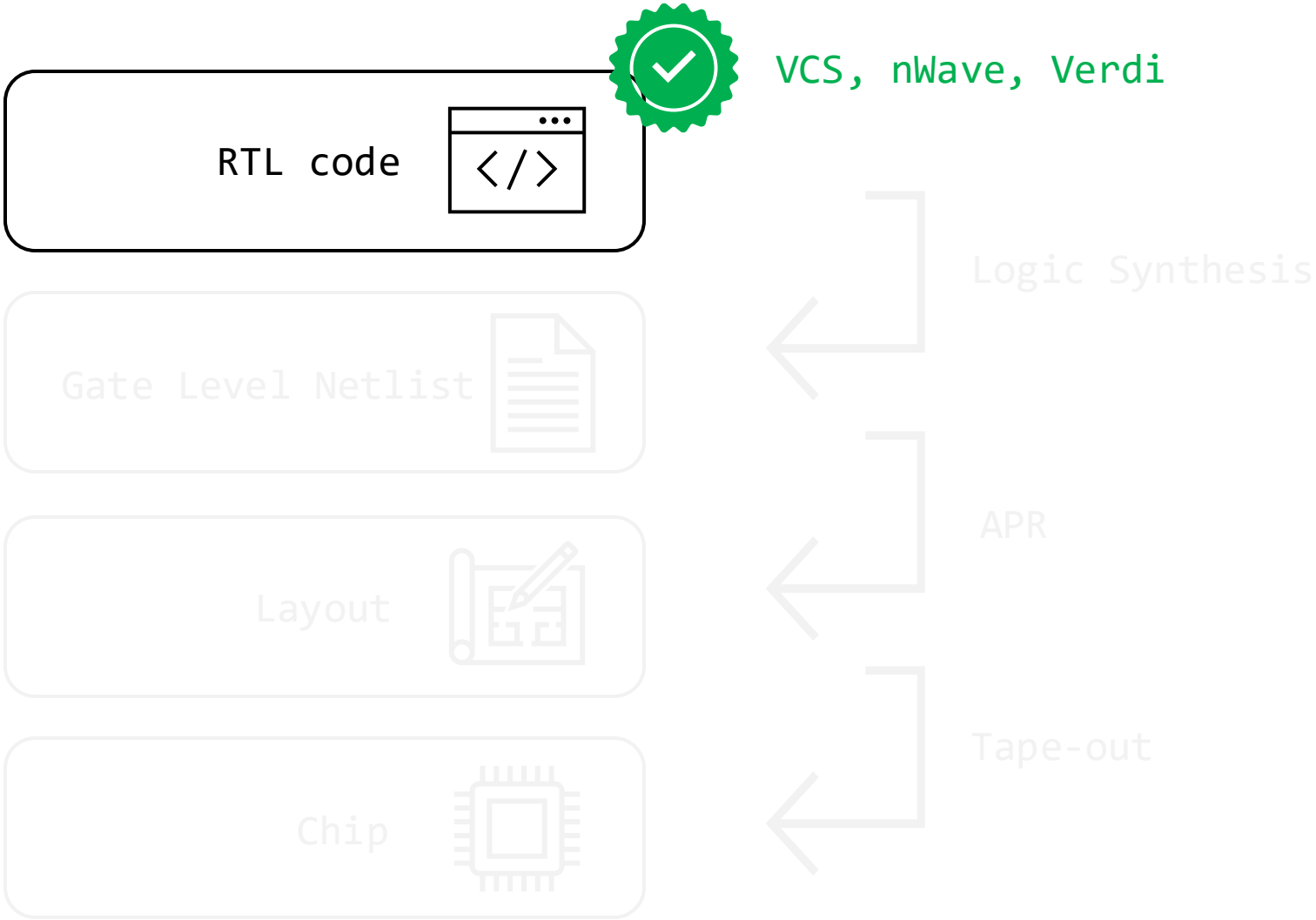
2025 HDL/ALU

1. Design Your Circuit by code



2. Pre-Simulation



VCS simulation & nWave

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2025 ALU / HDL

VCS Simulation

- 利用Testbench提供測試訊號
- 模擬電路訊號與運作狀況

- 輸入檔案
  - Testbench
  - Your Design
- 輸出檔案
  - fsdb Waveform file

VCS Pre-Simulation

1. 輸入tcsh

```
[m123040033@DTrump pre_sim]$ tcsh
set vcs version: 2023.12 (default)
set verdi version: 2023.12 (default)
set synthesis version: 2023.12/ (default)
set lc version: 2023.12/ (default)
set formality version: 2023.12/ (default)
set primetime version: 2023.12/ (default)
set INNOVUS version: INNOVUS_21.17.000 (default)
```

2. 編輯pre_sim.sh

testbench

Your Design

Output Waveform
可以修改成自己喜歡的檔名

sim.sh Workspace Trust

C: > Users > ivan > AppData > Roaming > MobaXterm > slash > RemoteFiles > 133708_2_2 > sim.sh

```
#!/bin/tcsh

vcs -R \
/home/m123040033/HDL/pre_sim/TB.v \
/home/m123040033/HDL/rtl/adder_behavior_reg.v\
/home/m123040033/HDL/rtl/adder_behavior.v\
/home/m123040033/HDL/rtl/adder_dataflow_reg.v\
/home/m123040033/HDL/rtl/adder_dataflow.v\
/home/m123040033/HDL/rtl/adder_structure_reg.v\
/home/m123040033/HDL/rtl/adder_structure.v\
/home/m123040033/HDL/rtl/D_FF_32.v\
/home/m123040033/HDL/rtl/D_FF.v\
/home/m123040033/HDL/rtl/FullAdder.v \
+full64 \
+access+r +vcs+fsdbon +fsdb+mda +fsdbfile+6adder.fsdb
```

建議
1. 使用絕對路徑
2. 自己打路徑的內容

VCS Pre-Simulation

3. source pre_sim.sh

```
[m123040033@Elden pre_sim]$ source sim.sh
*** Using c compiler gcc instead of cc ...
                    Chronologic VCS (TM)
          Version V-2023.12 -- Fri Sep 20 16:06:47 2024

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                      https://solvnetplus.synopsys.com)

Parsing design file '/home/m123040033/HDL/pre_sim/TB.v'
Parsing design file '/home/m123040033/HDL/rtl/adder_behavior_reg.v'
Parsing design file '/home/m123040033/HDL/rtl/adder_behavior.v'
Parsing design file '/home/m123040033/HDL/rtl/adder_dataflow_reg.v'
Parsing design file '/home/m123040033/HDL/rtl/adder_dataflow.v'
Parsing design file '/home/m123040033/HDL/rtl/adder_structure_reg.v'
Parsing design file '/home/m123040033/HDL/rtl/adder_structure.v'
Parsing design file '/home/m123040033/HDL/rtl/D_FF_32.v'
Parsing design file '/home/m123040033/HDL/rtl/D_FF.v'
Parsing design file '/home/m123040033/HDL/rtl/FullAdder.v'
Top Level Modules:
       HW1_TB
TimeScale is 1 ns / 100 ps
Starting vcs inline pass ...

3 modules and 0 UDP read.
recompiling module HW1_TB
recompiling module adder_structure
recompiling module FA
All of 3 modules done
```
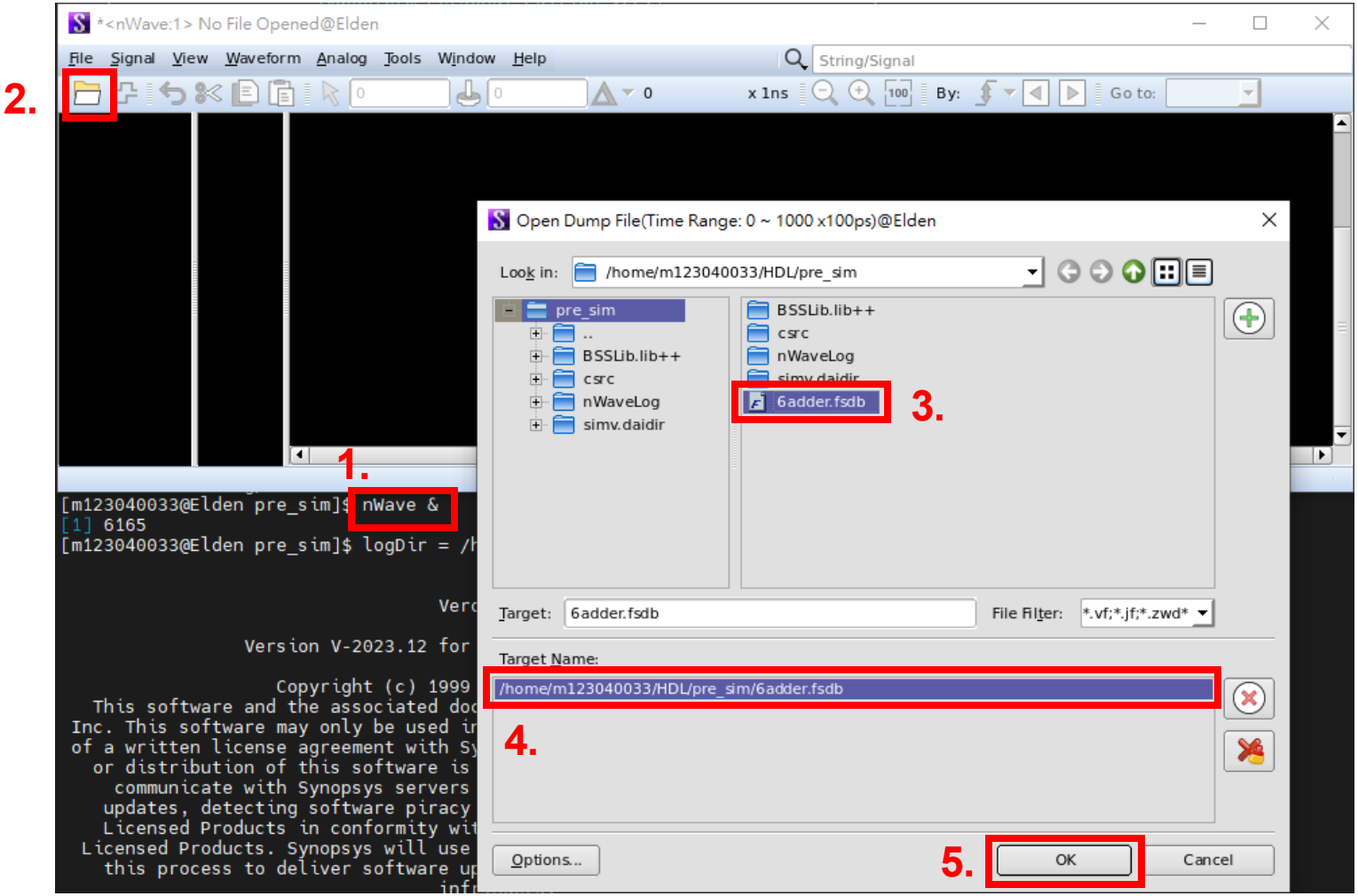
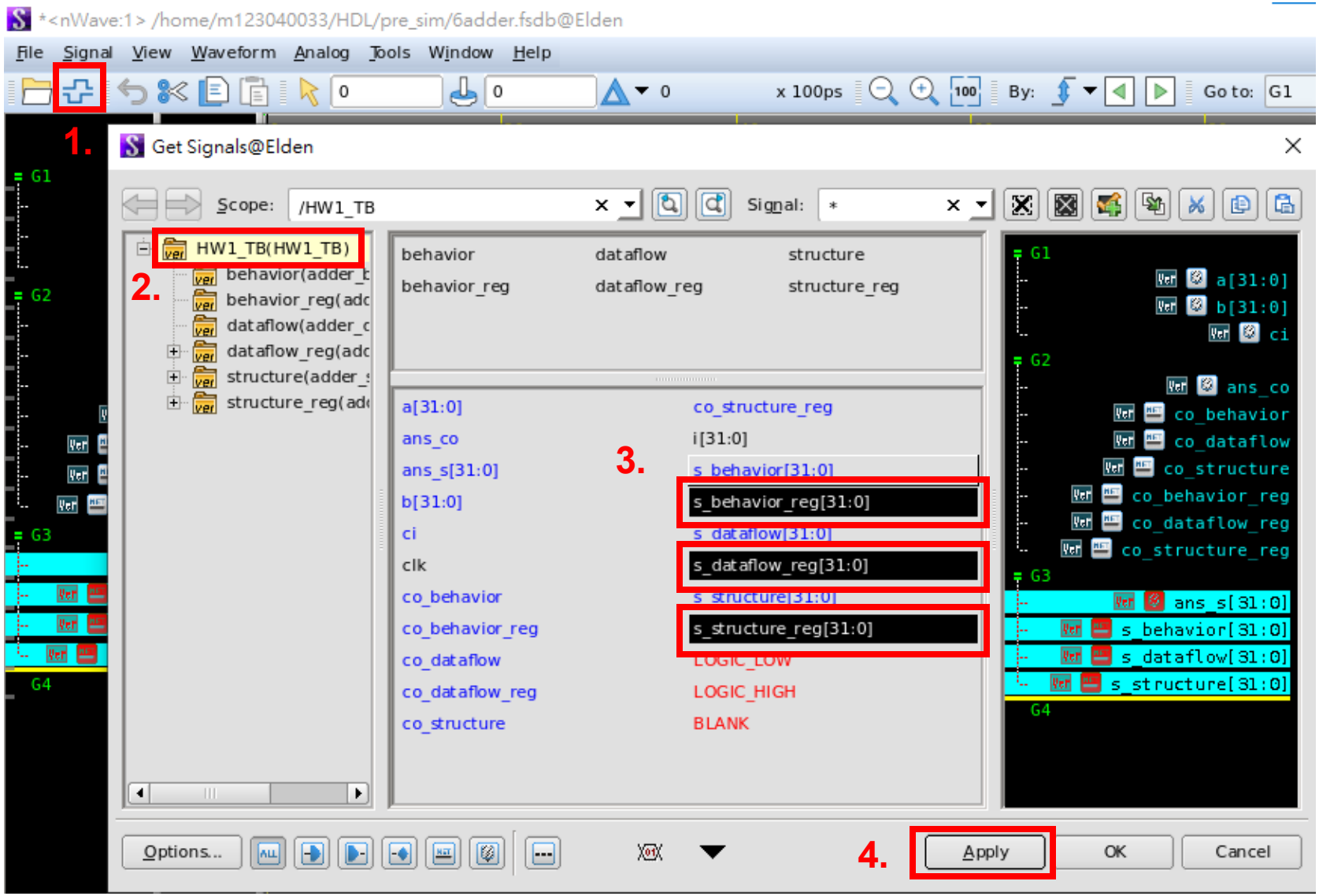

```
3 modules and 0 UDP read.
recompiling module HW1_TB
recompiling module adder_structure
recompiling module FA
All of 3 modules done
rm -f _cuarc*.so _csrc*.so pre_vcsobj_*.so share_vcsobj_*.so
if [ -x ../simv ]; then chmod a-x ../simv; fi
g++  -o ../simv      -m32 -m32 -rdynamic  -Wl,-rpath='$ORIGIN'/simv.daidir -Wl,-rpath=./simv.da
idir -Wl,-rpath=/usr/cad/synopsys/vcs/2023.12/linux/lib -L/usr/cad/synopsys/vcs/2023.12/linux/l
ib  -Wl,-rpath-link=./   objs/amcQw_d.o   _4401_archive_1.so _prev_archive_1.so   SIM_l.o
 rmapats_mop.o rmapats.o rmar.o rmar_nd.o  rmar_llvm_0_1.o rmar_llvm_0_0.o            -lvirsim
-lerrorinf -lsnpsmalloc -lvfs    -lvcsnew -lsimprofile -luclinative /usr/cad/synopsys/vcs/2023.
12/linux/lib/vcs_tls.o   -Wl,-whole-archive  -lvcsucli    -Wl,-no-whole-archive         _vcs_pli
_stub_.o   /usr/cad/synopsys/vcs/2023.12/linux/lib/vcs_save_restore_new.o /usr/cad/synopsys/ver
di/2023.12/share/PLI/VCS/LINUX/pli.a /usr/cad/synopsys/vcs/2023.12/linux/lib/ctype-stubs_32.a -
ldl  -lc -lm -lpthread -ldl
../simv up to date
Info: [VCS_SAVE_RESTORE_INFO] ASLR (Address Space Layout Randomization) is detected on the mach
ine. To enable $save functionality, ASLR will be switched off and simv re-executed.
Please use '-no_save' simv switch to avoid re-execution or '-suppress=ASLR_DETECTED_INFO' to su
ppress this message.
Chronologic VCS simulator copyright 1991-2023
Contains Synopsys proprietary information.
Compiler version V-2023.12; Runtime version V-2023.12;  Sep 20 16:07 2024
*Verdi* Loading libsscore_vcs202312.so
FSDB Dumper for VCS, Release Verdi_V-2023.12, Linux, 11/15/2023
(C) 1996 - 2023 by Synopsys, Inc.
*Verdi* FSDB WARNING: The FSDB file already exists. Overwriting the FSDB file may crash the pro
grams that are using this file.
*Verdi* : Create FSDB file '6adder.fsdb'
*Verdi* : Begin traversing the scopes, layer (0).
*Verdi* : Enable +mda dumping.
*Verdi* : End of traversing.
$finish called from file "/home/m123040033/HDL/pre_sim/TB.v", line 35.
$finish at simulation time                 1000
           V C S   S i m u l a t i o n   R e p o r t
Time: 100000 ps
CPU Time:      0.660 seconds;       Data structure size:   0.0Mb
Fri Sep 20 16:07:10 2024
CPU time: .558 seconds to compile + .499 seconds to elab + .412 seconds to link + .722 seconds
in simulation
```

nWave – Select File

- nWave & //&是在背景執行

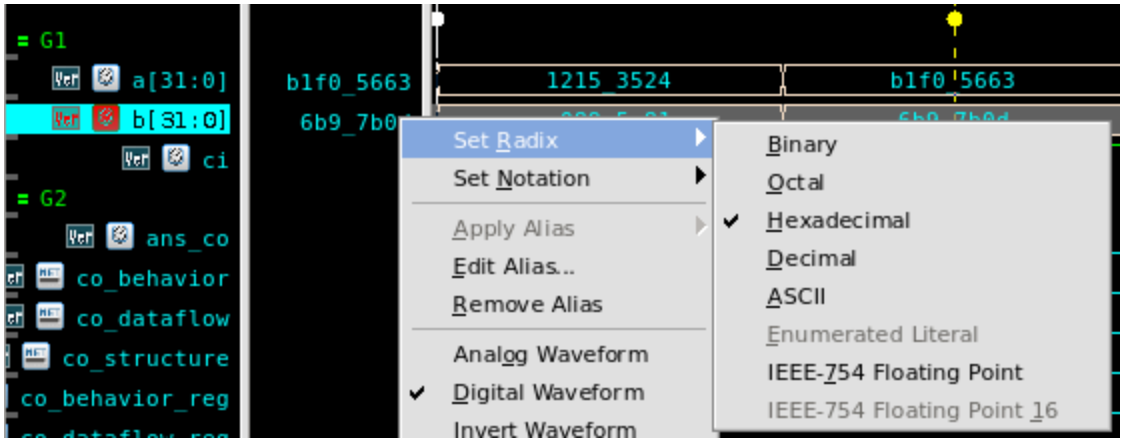


nWave – Select Signal



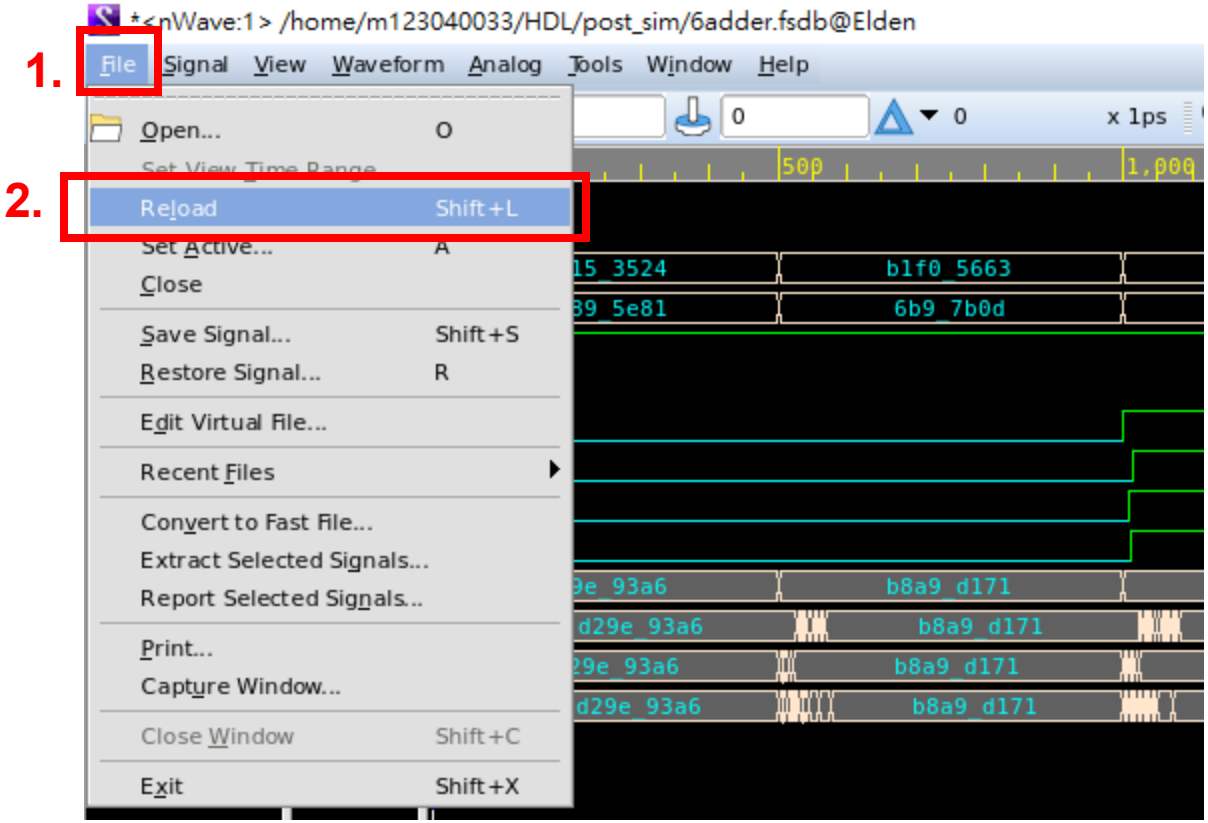
nWave - 常用操作

- 縮放 ： Ctrl + 滑鼠滾輪
- 調整信號位置： 按住滑鼠中鍵拖曳
- 調整數值表示： 右鍵 > Set Radix



nWave - Reload

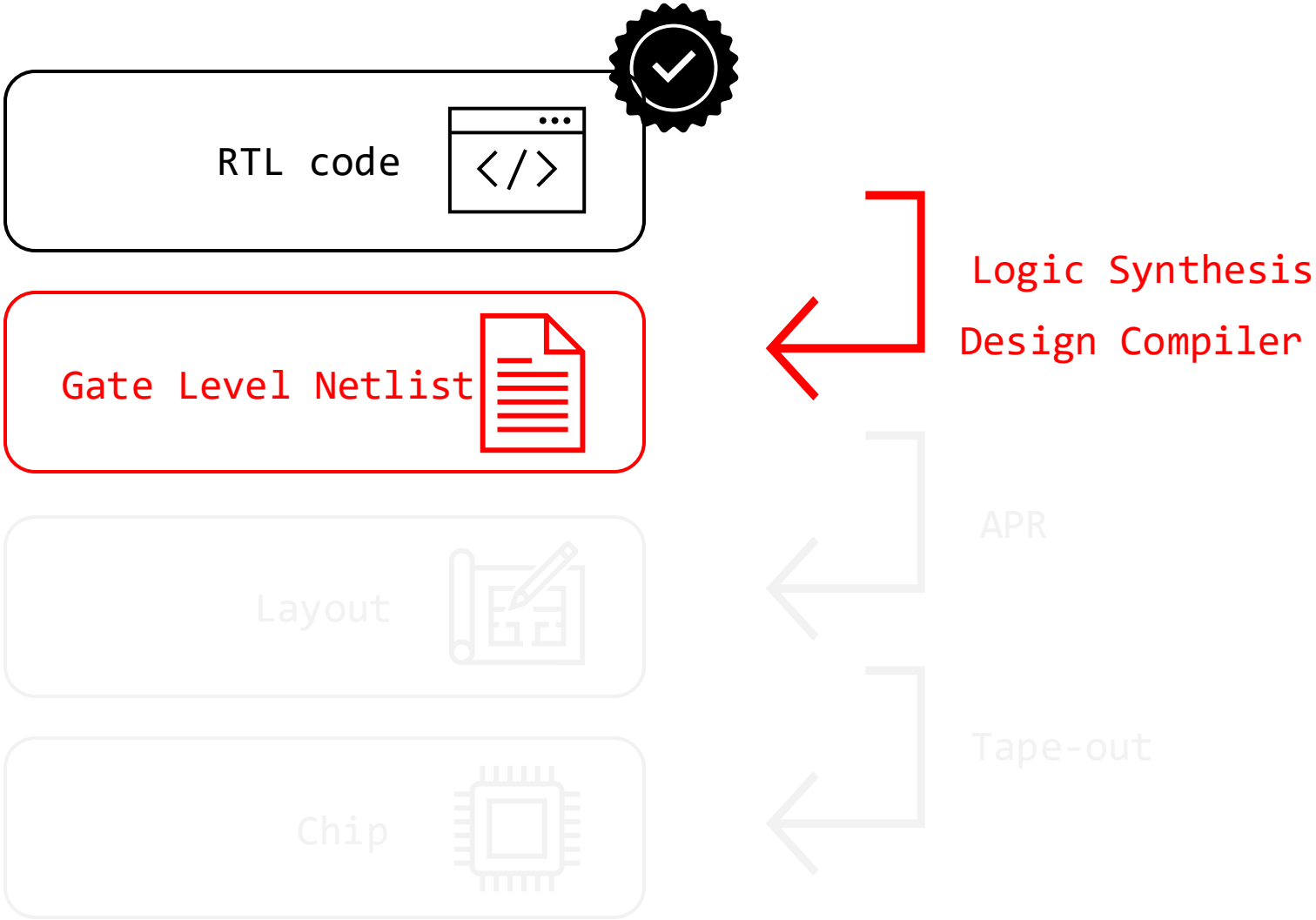
- 重新執行 VCS 模擬後，不必重開nWave和fsdb



Verdi

- Debug 軟體，有需要的同學請自行到 Youtube 觀看。
- https://youtu.be/wTpzvZs4xZU

3. Logic Synthesis



Design Compiler

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2025 ALU / HDL

所需檔案

- RTL code (你的程式碼，不包含testbench)
- 執行Design compiler 之腳本 (作業檔案提供)
- TSMC ADFP Cell Library File (伺服器內部)
- EDA Tools (伺服器內部)

腳本說明

- dc.tcl

```
#設定ADFP(16nm)製程路徑
set search_path      "/cad/CBDK/ADFP/Executable_Package/Collaterals/IP/stdcell/N16ADFP_StdCell/CCS/  $search_path"
#設定ADFP(16nm)製程路徑檔，如果有memory compiler的檔案db檔的路徑，記得在這邊設定
set target_library   "N16ADFP_StdCellss0p72vm40c_ccs.db N16ADFP_StdCellff0p88v125c_ccs.db"

set link_library     "* $target_library dw_foundation.sldb"
set symbol_library   "tsmc040.sdb generic.sdb"
set synthetic_library "dw_foundation.sldb"
set hdlin_translate_off_skip_text "TRUE"
set edifout_netlist_only "TRUE"
set verilogout_no_tri true
set hdlin_enable_presto_for_vhdl "TRUE"
set sh_enable_line_editing true
set sh_line_editing_mode emacs
history keep 100
alias h history
```

腳本說明

- 修改檔案存放位置、top module name及clock

```
#Path_Top:Verilog放置的位置
#Path_Syn:合成後report.txt檔案要放置的根位置，需自行在目錄下創建名為dc_out_file之資料夾
#Dump_file_name:合成後產生檔案之名字
set Path_Top        "./"
set Path_Syn        "../structure_reg/"
set Dump_file_name "adder_structure_reg"
#設定Top module 名稱，需跟自行設計之電路的top module name相同
set Top             "adder_structure_reg"
#Specify Clock，clock名需和top module中clk port相同
set Clk_pin         "clk"
set Clk_period      "0.49"
```

- 將所有要合成的檔案之完整路徑放入 {} 內

```
# [ Read Design File ]
#如果設計有parameter設計，read_file指定不能用，需使用analyze + elaborate指令並自行更改路徑
# read_file -format verilog {/home/m103040049/HDL_HW/multiplier.v}
# current_design $Top
analyze -format verilog {
        /home/m123040033/HDL/rtl/adder_structure_reg.v
        /home/m123040033/HDL/rtl/D_FF.v
        /home/m123040033/HDL/rtl/D_FF_32.v
        /home/m123040033/HDL/rtl/FullAdder.v
    }
elaborate $Top
current_design $Top
#檢查是否讀取成功
link
```

腳本說明

- 作業要求需合成 *delay-optimization*, *area-optimization*, and *in-between* 之三種電路，需修改31行中的period
  - 先將period設一個很大的值(假設10)進行合成，合成後report timing 發現最快可以跑2，則將period改設2再合一次，如果slack >= 0 (MET) 則 22 即為 *area-optimization* 之period
  - 接著改設period為一個很小的值(假設0)進行合成，合成後report timing發現slack = -0.5 (Violate)，則將period改設0.5再合一次，如果slack >= 0 (MET)則0.5即為 *delay-optimization* 之period，讓 slack 盡可能越接近0
  - 最後設period為(2+0.5)/2=1.25進行合成，假設slack = -0.1(Violate)則設1.35再合一次，直到slack >= 0(MET)，最終period值即為*in-between* 電路之period

腳本說明

- 作業要求需合成 *delay-optimization*, *area-optimization*, and *in-between*之三種電路，需修改31行中的period

```
21  #Path_Top:Verilog放置的位置
22  #Path_Syn:合成後report.txt檔案要放置的根位置，需自行在目錄下創建名為dc_out_file之資料夾
23  #Dump_file_name:合成後產生檔案之名字
24  set Path_Top        "./"
25  set Path_Syn        "../structure_reg/"
26  set Dump_file_name "adder_structure_reg"
27  #設定Top module 名稱，需跟自行設計之電路的top module name相同
28  set Top             "adder_structure_reg"
29  #Specify Clock，clock名需和top module中clk port相同
30  set Clk_pin         "clk"
31  set Clk_period      "0.49"
```

腳本說明

- 如果電路中無clk設計，使用50、51行，註解54~57行

```
49  # [ Setting Clock Constraints, Combinational Circurt USED ]
50  set_max_delay $Clk_period  -from [all_inputs] -to [all_outputs]
51  create_clock -name $Clk_pin -period $Clk_period
52
53  # [ Setting Clock Constraints ]
54  # create_clock -name $Clk_pin -period $Clk_period [get_ports $Clk_pin]
55  # set_fix_hold                                    [get_clocks $Clk_pin]
56  # set_dont_touch_network                              [get_clocks $Clk_pin]
57  # set_ideal_network                               [get_ports $Clk_pin]
```

- 如果電路中有clk設計，使用54~57行，註解50、51行

```
49  # [ Setting Clock Constraints, Combinational Circurt USED ]
50  # set_max_delay $Clk_period  -from [all_inputs] -to [all_outputs]
51  # create_clock -name $Clk_pin -period $Clk_period
52
53  # [ Setting Clock Constraints ]
54  create_clock -name $Clk_pin -period $Clk_period [get_ports $Clk_pin]
55  set_fix_hold                                    [get_clocks $Clk_pin]
56  set_dont_touch_network                          [get_clocks $Clk_pin]
57  set_ideal_network                               [get_ports $Clk_pin]
```

腳本說明

- 設定合成時命名規則(不須修改)
- 輸出timing、area和power report，輸出路徑可至25行更改
- 產生合成後.v檔、ddc、sdf和sdc 檔，產生之名稱可至26行更改

```
77  #Change Naming Rule
78  set bus_inference_style {%s[%d]}
79  set bus_naming_style {%s[%d]}
80  set hdlout_internal_busses true
81  change_names -hierarchy -rule verilog
82  define_name_rules name_rule -allowed "A-Z a-z 0-9 _" -max_length 255 -type cell
83  define_name_rules name_rule -allowed "A-Z a-z 0-9 _[]" -max_length 255 -type net
84  define_name_rules name_rule -map {{"\\*cell\\*""cell"}}
85  define_name_rules name_rule -case_insensitive
86  change_names -hierarchy -rules name_rule
87  remove_unconnected_ports -blast_buses [get_cells -hierarchical *]
88
89  report_timing -significant_digits 6 -sort_by group
90
91  #Report
92  report_timing -path full -delay max -significant_digits 6 -sort_by group > $Path_Syn/1.timing_report_${Dump_file_name}.txt
93  report_area -hier -nosplit > $Path_Syn/2.area_report_${Dump_file_name}.txt
94  report_power -analysis_effort low > $Path_Syn/3.power_report_${Dump_file_name}.txt
95
96  #Write out
97  write -hierarchy -format ddc -output $Path_Syn/${Dump_file_name}.ddc
98  write -format verilog -hierarchy -output $Path_Syn/${Dump_file_name}.v
99  write_sdf -version 2.1 -context verilog $Path_Syn/${Dump_file_name}.sdf
100 write_sdc $Path_Syn/${Dump_file_name}.sdc
```

使用Design Compiler合成

- 輸入dcnxt_shell –f dc.tcl ，即開始執行tcl檔、進行合成

```
[m103040049@devotion ~/ALU_HW1]$ dc_shell -f dc.tcl

                         Design Compiler Graphical
                               DC Ultra (TM)
                                DFTMAX (TM)
                            Power Compiler (TM)
                               DesignWare (R)
                               DC Expert (TM)
                             Design Vision (TM)
                             HDL Compiler (TM)
                             VHDL Compiler (TM)
                                DFT Compiler
                             Design Compiler(R)

                 Version O-2018.06 for linux64 - May 21, 2018

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 of a written license agreement with Synopsys, Inc. All other use, reproduction,
            or distribution of this software is strictly prohibited.
Initializing...
set Company             "VLSI5015"
VLSI5015
set Designer            "default"
default
#設定40nm製程路徑
set search_path      "/EDA_Tools/CBDK/CBDK_TSMC40_Arm_f2.0/CIC/SynopsysDC/db/sc9_b
```

Report Timing

```
stage2/adder/add_7/U1_23/CO (ADDF_X1M_A9TR)              0.08       2.07 r
stage2/adder/add_7/U1_24/CO (ADDF_X1M_A9TR)              0.08       2.15 r
stage2/adder/add_7/U1_25/CO (ADDF_X1M_A9TR)              0.08       2.24 r
stage2/adder/add_7/U1_26/CO (ADDF_X1M_A9TR)              0.08       2.32 r
stage2/adder/add_7/U1_27/CO (ADDF_X1M_A9TR)              0.08       2.40 r
stage2/adder/add_7/U1_28/S (ADDF_X1M_A9TR)               0.18       2.58 f
stage2/adder/add_7/SUM[28] (adder_DW01_add_0)            0.00       2.58 f
stage2/adder/sign_d (adder)                              0.00       2.58 f
stage2/conversion_sign/sign_d (conversion_sign)          0.00       2.58 f
stage2/conversion_sign/U34/Y (BUFH_X1M_A9TR)             0.08       2.67 f
stage2/conversion_sign/U5/Y (INV_X1M_A9TR)               0.03       2.70 r
stage2/conversion_sign/U3/Y (BUFH_X1M_A9TR)              0.08       2.78 r
stage2/conversion_sign/U15/Y (AO22_X1M_A9TR)             0.09       2.87 r
stage2/conversion_sign/mantissa_out[9] (conversion_sign)
                                                         0.00       2.87 r
stage2/mantissa_unsigned_reg_9_/D (DFFRPQ_X2M_A9TR)      0.00       2.87 r
data arrival time                                                   2.87
```

表格 Delay 欄位需要填入的部分

```
clock clk (rise edge)                                    3.00       3.00
clock network delay (ideal)                              0.00       3.00
stage2/mantissa_unsigned_reg_9_/CK (DFFRPQ_X2M_A9TR)
                                                         0.00       3.00 r
library setup time                                      -0.05       2.95
data required time                                                  2.95
-------------------------------------------------------------------------
data required time                                                  2.95
data arrival time                                                   -2.87
-------------------------------------------------------------------------
slack (MET)                                                         0.09
```

需要 >= 0 且有 slack (MET)

Report Area

```
****************************************
Report : area
Design : FLP_adder
Version: O-2018.06
Date   : Sat Oct  8 17:10:59 2022
****************************************

Library(s) Used:

    sc9_cln40g_base_rvt_ss_typical_max_0p81v_125c (File: /EDA_Tools/CBDK/CBDK_TSMC40_Arm_f2.0/CIC/SynopsysDC/db/sc9_ba

Number of ports:                          1545
Number of nets:                           3576
Number of cells:                          2029
Number of combinational cells:            1834
Number of sequential cells:                172
Number of macros/black boxes:                0
Number of buf/inv:                         409
Number of references:                        4

Combinational area:               2369.833178
Buf/Inv area:                      218.635199
Noncombinational area:             780.191961
Macro/Black Box area:                0.000000
Net Interconnect area:      undefined  (Wire load has zero net area)

Total cell area:                  3150.025139
Total area:                 undefined

Hierarchical area distribution
------------------------------
```

Combinational Logic （表格 **CL** 欄位）

Sequential Logic （表格 **SL** 欄位）

Toral Area（表格 **Total** 欄位）

Report Power

```
Design        Wire Load Model            Library
------------------------------------------------
FLP_adder              Zero              sc9_cln40g_base_rvt_ss_typical_max_0p81v_125c


Global Operating Voltage = 0.81
Power-specific unit information :
    Voltage Units = 1V
    Capacitance Units = 1.000000pf
    Time Units = 1ns
    Dynamic Power Units = 1mW    (derived from V,C,T units)
    Leakage Power Units = 1uW


  Cell Internal Power  = 287.2819 uW   (86%)
  Net Switching Power  =  46.9000 uW   (14%)
                         ---------
Total Dynamic Power    = 334.1819 uW  (100%)

Cell Leakage Power     =  29.8890 uW


                 Internal         Switching           Leakage            Total
Power Group      Power            Power               Power              Power   (   %    )  Attrs
--------------------------------------------------------------------------------------------------
io_pad             0.0000           0.0000            0.0000            0.0000  (   0.00%)
memory             0.0000           0.0000            0.0000            0.0000  (   0.00%)
black_box          0.0000           0.0000            0.0000            0.0000  (   0.00%)
clock_network      0.0000           0.0000            0.0000            0.0000  (   0.00%)
register           0.2055        4.9244e-03           7.7363            0.2182  (  59.93%)
sequential         0.0000           0.0000            0.0000            0.0000  (   0.00%)
combinational  8.1768e-02        4.1976e-02          22.1417            0.1459  (  40.07%)
--------------------------------------------------------------------------------------------------
Total              0.2873 mW     4.6900e-02 mW        29.8780 uW         0.3641 mW
```

請轉換單位: uW → W (表格要求)

Dynamic Power（表格 **Dynamic** 欄位）

Leakage Power（表格 **Leakage** 欄位）

Dynamic + Leakage = Total（表格 **Total** 欄位）

VCS Post-Simulation

```
#!/bin/tcsh

vcs -R -error=noMPD \
/home/m123040033/HDL/post_sim/TB.v \
/home/m123040033/HDL/behavior/adder_behavior.v \
/home/m123040033/HDL/behavior_reg/adder_behavior_reg.v \
/home/m123040033/HDL/dataflow/adder_dataflow.v \
/home/m123040033/HDL/dataflow_reg/adder_dataflow_reg.v \
/home/m123040033/HDL/structure/adder_structure.v \
/home/m123040033/HDL/structure_reg/adder_structure_reg.v \
/cad/CBDK/ADFP/Executable_Package/Collaterals/IP/stdcell/N16ADFP_StdCell/VERILOG/N16ADFP_StdCell.v \
+full64 \
+access+r +vcs+fsdbon +fsdb+mda +fsdbfile+6adder.fsdb +neg_tchk
```

testbench

Gate Level Netlist

Standard Cell

Output Waveform
可以修改成自己喜歡的檔名

VCS Post-Simulation - TB

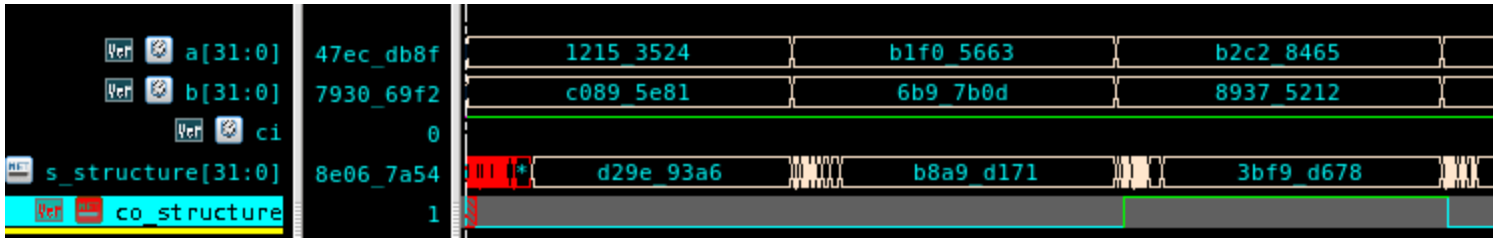
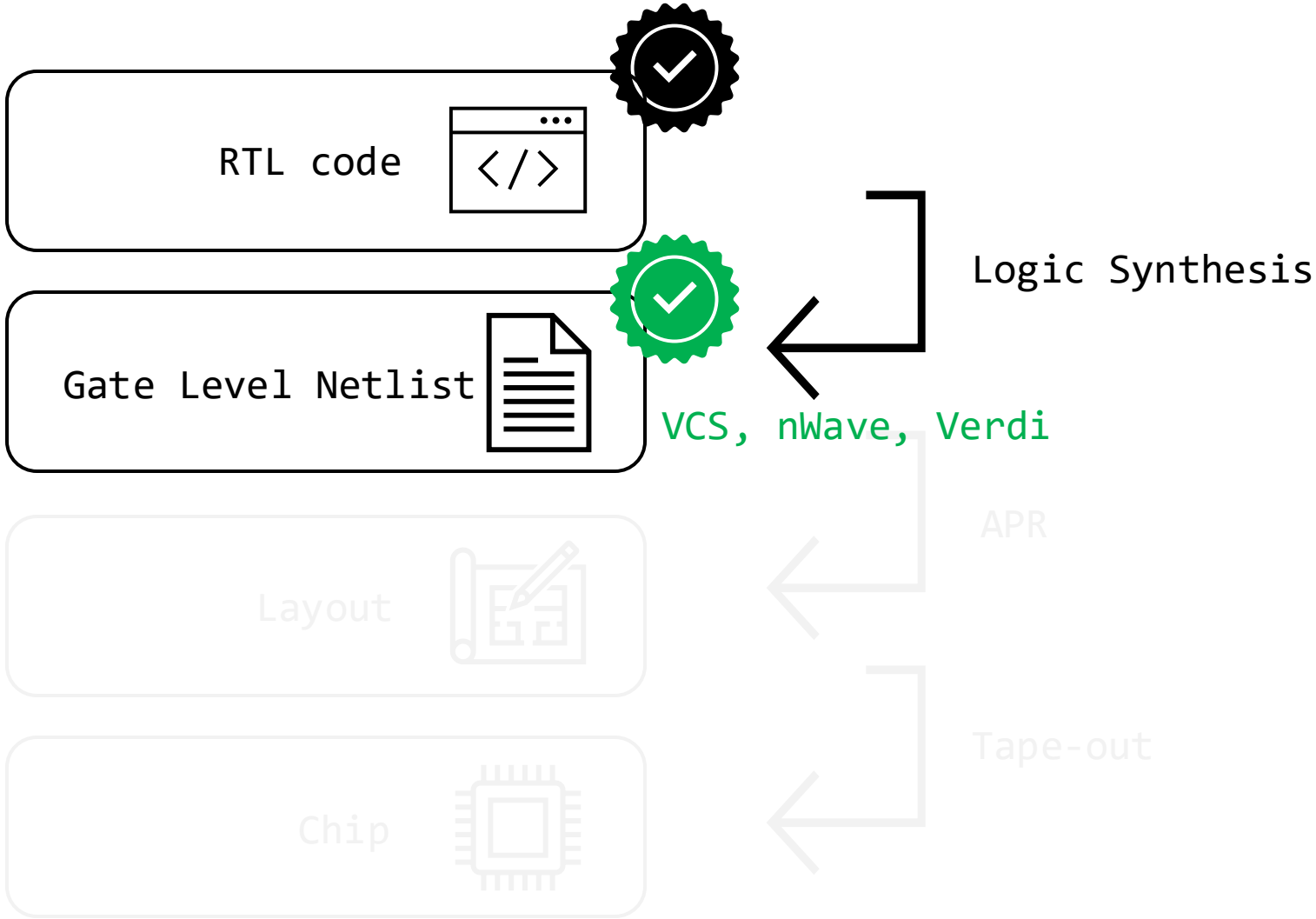
需要在 TB 內加入 Standard Delay Format （SDF）

```
adder_behavior behavior(s_behavior, co_behavior, a, b, ci);
adder_behavior_reg behavior_reg(s_behavior_reg, co_behavior_reg, a, b, ci, clk);
adder_dataflow dataflow(s_dataflow, co_dataflow, a, b, ci);
adder_dataflow_reg dataflow_reg(s_dataflow_reg, co_dataflow_reg, a, b, ci, clk);
adder_structure structure(s_structure, co_structure, a, b, ci);
adder_structure_reg structure_reg(s_structure_reg, co_structure_reg, a, b, ci, clk);
```

SDF File path

這邊要和呼叫的 Module Instance name 一樣

```
initial $sdf_annotate("/home/m123040033/HDL/behavior/adder_behavior.sdf", behavior);
initial $sdf_annotate("/home/m123040033/HDL/behavior_reg/adder_behavior_reg.sdf", behavior_reg);
initial $sdf_annotate("/home/m123040033/HDL/dataflow/adder_dataflow.sdf", dataflow);
initial $sdf_annotate("/home/m123040033/HDL/dataflow_reg/adder_dataflow_reg.sdf", dataflow_reg);
initial $sdf_annotate("/home/m123040033/HDL/structure/adder_structure.sdf", structure);
initial $sdf_annotate("/home/m123040033/HDL/structure_reg/adder_structure_reg.sdf", structure_reg);
```



4. Post-Simulation